

Module 6

Virtualization and File System Management

Introduction to Virtual Machine

A virtual machine (VM) is a *virtual representation or emulation of a physical computer* that uses software instead of hardware to run programs and deploy applications.

VMs work by using virtualization, *a process of creating software-based or virtual versions of resources or applications.*

The fundamental idea behind a virtual machine is to abstract the hardware of a single computer into several different execution environments, thereby creating the illusion that each separate environment is running on its own private computer.

Introduction to Virtual Machine

A virtual machine is a computer file, typically called an image, that behaves like an actual computer.

It can run in a window as a separate computing environment, often to run a different operating system—or even to function as the user's entire computer experience—as is common on many people's work computers.

The virtual machine is partitioned from the rest of the system, meaning that the software inside a VM can't interfere with the host computer's primary operating system.

Introduction to Virtual Machine

Virtual machine implementations involve several components: At the base is the **host**, the underlying hardware system that runs the virtual machines.

The **virtual machine manager** (VMM) (also known as a **hypervisor**) creates and runs virtual machines by providing an interface that is identical to the host.

Each guest process is provided with a virtual copy of the host.

A single physical machine can thus run multiple operating systems concurrently, each in its own virtual machine.

Introduction to Virtual Machine

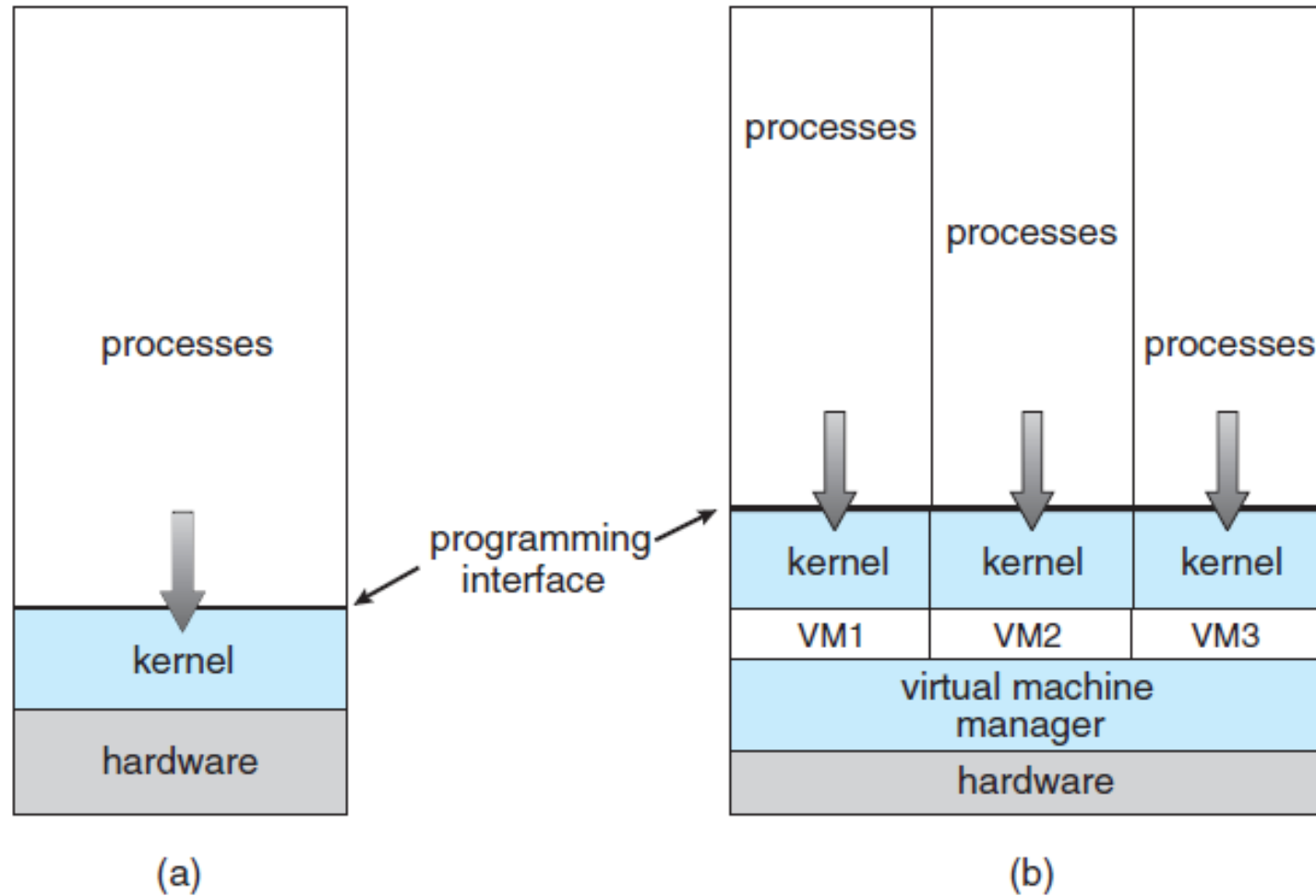


Figure: System models. (a) Nonvirtual machine. (b) Virtual machine

How do Virtual Machines work?

The hypervisor abstracts the host machine's physical resources, such as compute, memory, or storage, into a pool that can be provisioned and dynamically allocated to guest VMs as needed, providing more flexibility and increasing overall efficiency.

Types of Virtual Machines

There are two types of virtual machines:

1. Process VM:

A process virtual machine allows a single process to run as an application on a host machine, providing a platform-independent programming environment by masking the information of the underlying hardware or operating system.

It creates a **platform-independent environment** that lets an application or process run the same way on any platform.

Example: Java Virtual Machine

Types of Virtual Machines

There are two types of virtual machines:

2. System VM:

A system VM (sometimes called hardware virtual machines) **simulates a complete operating system**, allowing multiple OS environments to live on the same machine.

System VMs can run their own OS and applications, and a hypervisor monitors and distributes the physical host machine's resources between system VMs.

Example: VMware ESXi

Benefits of Virtual Machines

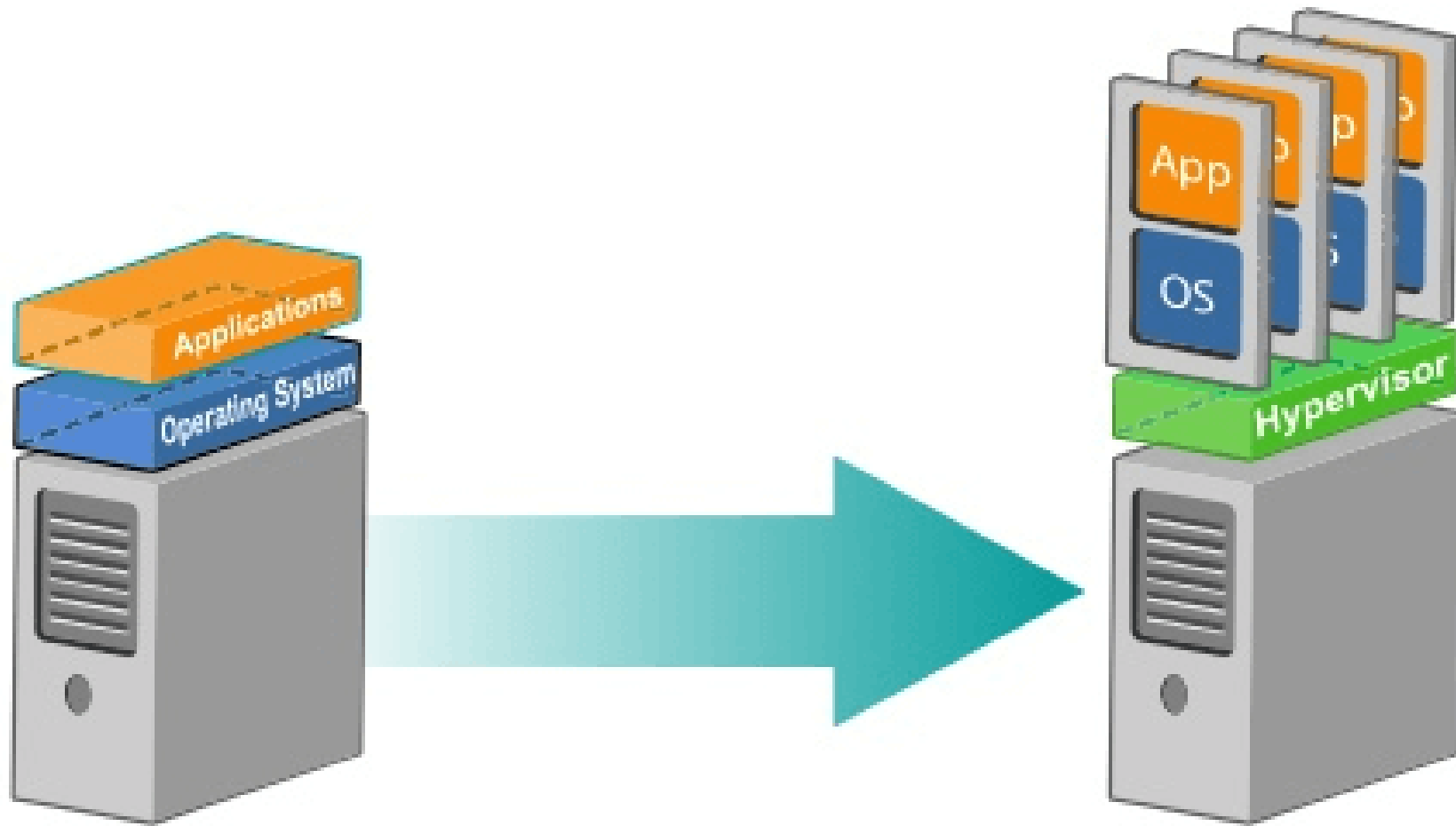
1. Resource utilization and improved ROI
2. Agility and speed
3. Portability - *VMs can even move between on-premises and cloud environments*
4. Flexibility – *create new environment on demand*
5. Reliability – *One VM virtually isolated from other VM's*
6. Security – *VM file can be scanned for Malware*
7. Sustainability - *reduce energy consumption*

Drawbacks of Virtual Machines

1. Performance issues – *hardware dependent*
2. Increased complexity – *complex to configure and manage*
3. **Single Point** of Failure – *rely on single physical computer*

Virtualization

Virtualization



Introduction to Virtualization

Virtualization is technology that can be used to create *virtual representations of servers, storage, networks, and other physical machines.*

It allows for more **efficient utilization of physical computer hardware**

It mimics the functions of physical hardware to run multiple virtual machines simultaneously on a single physical machine.

Virtualization uses software to create an abstraction layer over computer hardware that allows the hardware elements of a single computer (processors, memory, storage and more) to be divided into multiple virtual computers, commonly called virtual machines (VMs).

Each VM runs its own operating system (OS) and behaves like an independent computer even though it is running on just a portion of the actual underlying computer hardware.

Why Virtualization?

The goal of virtualization is to *support portability, improve efficiency, increase reliability, and shield the user from the complexity* of the system.

By using virtualization, you can *interact with any hardware resource* with greater flexibility.

Physical servers *consume electricity, take up storage space, and need maintenance*. You are often *limited by physical proximity and network design* if you want to access them.

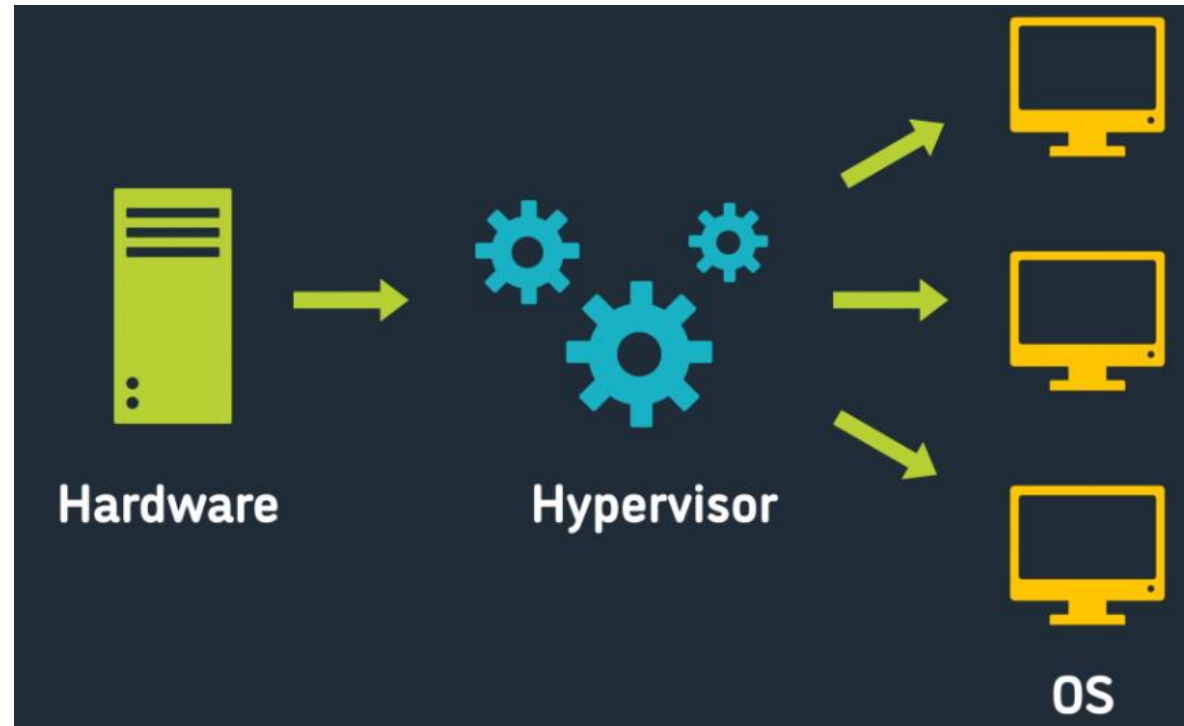
Virtualization removes all these limitations by abstracting physical hardware functionality into software. You can manage, maintain, and use your hardware infrastructure like an application on the web.

Benefits of Virtualization

- ❑ Resource efficiency - ability to run multiple VMs in one physical machine server
- ❑ Easier migration of application from one physical machine to another with minimal downtime
- ❑ Lower cost of ownership
- ❑ Simulate hardware and run programs in incompatible OS environments
- ❑ Faster provisioning

Hypervisor

The *hypervisor* is a **software component** that manages multiple virtual machines in a computer. It ensures that each virtual machine gets the allocated resources and does not interfere with the operation of other virtual machines.



What is hypervisor?

A **hypervisor**, also known as a virtual machine monitor or VMM, is software that creates and runs virtual machines (VMs).

A hypervisor allows one host computer to support multiple guest VMs by virtually sharing its resources, such as memory and processing.

The hypervisor allocates the underlying physical computing resources such as CPU and memory to individual virtual machines as required. Thus, it supports the optimal use of physical IT infrastructure.

There are two main types of hypervisor:

1. Type 1 or Native or “bare metal” hypervisors
2. Type 2 or Hosted or “embedded” hypervisors

Hypervisor types

Type 1 hypervisor

A type 1 hypervisor, or a bare metal hypervisor, interacts directly with the underlying machine hardware.

A bare metal hypervisor is **installed directly on the host machine's physical hardware**, not through an operating system. In some cases, a type 1 hypervisor is embedded in the machine's firmware.

The type 1 hypervisor negotiates directly with server hardware to allocate dedicated resources to VMs.

It can also flexibly share resources, depending on various VM requests.

Hypervisor types

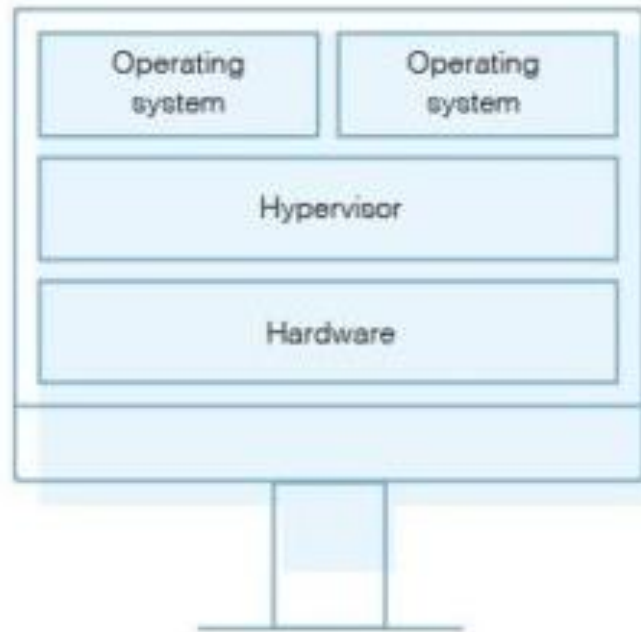
Type 2 hypervisor

A type 2 hypervisor, or hosted hypervisor, interacts with the underlying host machine hardware through the host machine's operating system.

- Install it on the machine, where it runs as an application.

The type 2 hypervisor negotiates with the operating system to obtain underlying system resources. However, the host operating system prioritizes its own functions and applications over the virtual workloads.

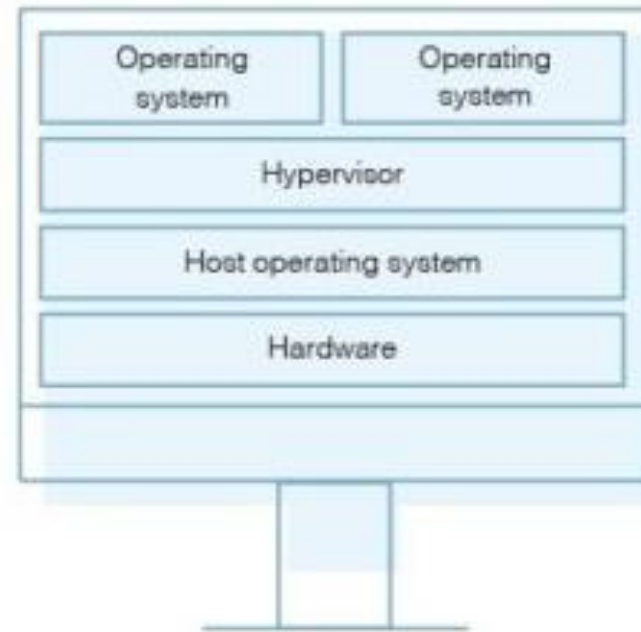
Two Types of Hypervisors



Native or Bare Metal

Runs directly on the hardware

- Isolates partitions
- Security features



Hosted or Embedded

Runs within and uses the host OS

- Ease to install and use
- Low cost

Examples

Type 1 Hypervisor

VMware hypervisors like vSphere, ESXi and ESX.

Microsoft Hyper-V.

Oracle VM Server.

Citrix Hypervisor.

Type 2 Hypervisor

Microsoft Virtual PC,

Oracle Virtual Box,

VMware Workstation,

Oracle Solaris Zones,

VMware Fusion,

Oracle VM Server for x86

Key differences: type 1 vs. type 2 hypervisors

While type 1 and type 2 hypervisors share the common goal to run and coordinate virtual machines (VMs), they have some significant variations.

Resource allocation

Type 1 hypervisors directly access underlying machine resources. They can implement their own custom resource allocation strategies to service their VMs.

Type 2 hypervisors negotiate resource allocation with the operating system, which makes the process slower and less efficient.

Ease of management

Managing a type 1 hypervisor and its VM configuration requires system administrator-level knowledge, as it's relatively complex.

In contrast, you can install and manage type 2 hypervisors as an application on an operating system. Even nontechnical users can operate them.

Performance

Type 1 hypervisors offer greater performance to their VMs. This is because they don't need to negotiate resources with the operating system or travel through the operating system layer. The type 1 hypervisor offers dedicated underlying resources without any negotiation required.

Type 2 hypervisors must only use the resources that the operating system is willing to provide.

Isolation

Type 1 hypervisors offer a greater degree of isolation for each virtual environment. There's no shared layer like there is with the operating system for a type 2 hypervisor. This makes virtual machines running on the type 1 hypervisor inherently more secure. However, updating and patching your virtual machine operating systems is a critical security activity.